

Programmed iteration method for airline network optimization

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Airline network optimization. Static case.

$$Revenue = \langle f, x \rangle \rightarrow \max,$$

$$Ax \leq c(t),$$

$$0 \leq x \leq \mu.$$

where

- ▶ f_i is ODIF price.
- ▶ x_i is allocation for ODIF.
- ▶ A is product matrix.
- ▶ μ is ODIF demand.

Airline network optimization. Dynamic case.

Network optimized on finite horizon $[t_0, T]$.

$$Revenue = \int_{t_0}^T \langle f(\tau), x(\tau) \rangle d\tau \rightarrow \max,$$

$$A(t)x(t) \leq c(t),$$

$$0 \leq x(t) \leq \mu(t).$$

Given dynamic product matrix $A(t)$, capacities $c(t)$, prices $f(t)$ and demand $\mu(t)$ find dynamic allocations $x(t)$ maximizing revenue.

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Airline network optimization. Simplified case.

Simplification assumptions:

- ▶ Product matrix $A(t) = A$ is constant.
- ▶ Capacity vector $c(t) = c$ is constant.
- ▶ Price vector $f(t) = f$ is constant.

Then

$$\text{Revenue} = \int_{t_0}^T \langle f, x(\tau) \rangle d\tau \rightarrow \max,$$

$$Ax(t) \leq c,$$

$$0 \leq x(t) \leq \mu(t).$$

We solve this problem.

Programmed iteration method. Initial approximation.

Suppose constant allocation (control) x on $[t_0, T]$ is used.

$$\langle f, x \rangle (T - t_0) \rightarrow \max,$$

$$Ax \leq c,$$

$$0 \leq x \leq \min_{t \in [t_0, T]} \mu(t).$$

Classic Linear Programming (LP) problem.

Define $X(t) = \{x | Ax \leq c, 0 \leq x \leq \min_{\tau \in [t, T]} \mu(\tau)\}$.

Initial approximation

$$w^{(0)}(t) = \max_{x \in X(t)} \langle f, x \rangle (T - t).$$

Interpretation: maximum revenue achievable on $[t, T]$ using some constant allocations.

Programmed iteration method. Next iterations

Suppose we allowed to use one switch of constant allocation policy.

The maximum achievable revenue is then

$$w^{(1)}(t) = \max_{\tau \in [t, T]} \max_{x \in X(\tau)} \left[\langle f, x \rangle (\tau - t) + w^{(0)}(\tau) \right]$$

...and so on

$$w^{(n)}(t) = \max_{\tau \in [t, T]} \max_{x \in X(\tau)} \left[\langle f, x \rangle (\tau - t) + w^{(n-1)}(\tau) \right]$$

Taking a limit

$$w^{(n)}(\cdot) \longrightarrow w^*(\cdot).$$

$w^*(\cdot)$ is maximum revenue.

Computations

To find $w^{(0)}(\cdot)$ for each $t \in [t_0, T]$

- Solve LP.

To find $w^{(n)}(\cdot)$ for each $t \in [t_0, T]$

- For each $\tau \in [t, T]$ solve LP.
- Maximize by τ over $[t, T]$.

Suppose time is discretized with n steps.

For each iteration we require to solve n^2 LPs and n maximization of non-linear function.

For given demand vector $\mu(t)$ closed-form solutions (perhaps) possible.

Further research

- ▶ Get rid of simplification assumptions.
- ▶ Stochastic version (demand is stochastic process).
- ▶ Overbooking version (allow sell more than capacity).
- ▶ Game-theoretic version (two or more players).